

Cross-Sectional Reproduction of the SPX Implied-Vol Smile, the VIX Term Structure, and the VIX-Option Smile by the SP500 Feedback-Layer ESN (2026-02-19 Snapshot)

Companion note to the ESN A2-981003 joint-calibration family
(feedback-layer/ α version)

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Abstract

The joint-calibration note family fits the ESN-A2 architecture to 5 time-series stylised facts of daily returns – roughness, leverage, the Taylor effect, the Zumbach effect, and excess kurtosis – none of which touch option prices at all. This note asks a different, cross-sectional question: turned into an option-pricing engine (the simulator is already a discrete-time exponential martingale under its own measure – Proposition 1 in Section 3 – so no risk-premium adjustment is needed), does the SP500 calibration ($H = 0.386$, `rough_scale` = 0.672, $\lambda_{lo} = 10^{-5}$, $\lambda_{hi} = 4.8809$, $m_1 = +0.1745$, $\alpha = 0.220$) reproduce the actual SPX implied-vol smile, the VIX term structure, and the VIX-option smile observed on 2026-02-19 – and can a genuine cross-sectional calibration close the gap? As calibrated for the time series alone, no: the model overstates level by $\approx 1.8\times$ on both surfaces (30-day VIX = 36.0 vs. market = 19.6) and shows a real shape mismatch on both smiles. Section 3.3 fits all 8 free parameters (H , `rough_scale`, λ_{lo} , λ_{hi} , m_1 , m_2 , α , Δb_0) jointly against a vega-weighted implied-vol objective, sharpened toward the money with an explicit boost on $m \geq 1$ points, against *all 6 quoted expiries at once*: SPX near and next term, plus VIX near, next, 118d, and 153d – calibrating directly on the mid of each strike’s put and call implied vol (1 point per strike, not 2 independent targets) rather than the raw put and call quotes separately. The result: SPX near/next weighted RMSE 0.029/0.026, VIX near/next/118d/153d 0.29/0.08/0.09/0.11 (confirm-quality, averaged over 2 independent Monte-Carlo seeds – Table 4). The SPX smile now coincides with the market on both the $m < 1$ and $m \geq 1$ sides at comparable RMSE (previously $m \geq 1$ was $\approx 3.5\times$ worse, now $\approx 1.4\times$); the VIX smile tracks the market closely across most of its quoted range at 55d and 118d, with a real but smaller residual gap at 27d and 153d – an honest limit of the shared 8-parameter architecture fitting 4 VIX maturities’ own term structure at once, not a fitting failure. Priced at 2 further, entirely held-out SPX maturities (Section 3.3.3), the fit does not generalize uniformly: the SPX put wing undershoots increasingly with maturity – an honest limit of a 2-maturity SPX calibration set, not a contradiction of the results above. The VIX-option test itself needs a genuinely different, nested Monte-Carlo algorithm – detailed in full in Section 3.2 – since VIX options price the model-free variance rate observed at a *future* date, not the underlying level. All tests use the real `esn_base.py/joint_calibration.py` engine end to end, with no modification.

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1 Market data

2 option-chain snapshots, both dated 2026-02-19: SPX index options (11 expiries, 20 days to 5.8 years) and VIX options (4 expiries, 27 to 153 days). Quotes are mid implied volatilities (Black-76 convention), not dollar premia. The CBOE VIX methodology is applied literally,

$$\sigma^2 = \frac{2}{T} \sum_i \frac{\Delta K_i}{K_i^2} e^{RT} Q(K_i) - \frac{1}{T} \left[\frac{F}{K_0} - 1 \right]^2, \quad (1)$$

with $e^{RT}Q(K_i)$ built directly from each strike’s own mid-IV via undiscounted Black-76 (call for $K > K_0$, put for $K < K_0$, both averaged at K_0) – since $Q(K) = e^{-RT} \times \text{Black76}(K)$ exactly under this convention, R never needs to be estimated separately. Applied to the near-term (2026-03-20, $T_1 = 29\text{d}$) and next-term (2026-04-17, $T_2 = 57\text{d}$) SPX expiries and interpolated to a constant 30-day maturity (2-step CBOE procedure), this reproduces the file’s own quoted VIX spot (19.34) to within 1.5% (19.63 computed), validating the replication. The identical procedure on the VIX option chain gives a VVIX-style vol-of-VIX reading of 108.3% (used only as context; not part of either test below).

2 The model

This section documents the full architecture as implemented in `esn_base.py` – reservoir, z -bank, quadratic channel, feedback layer, and bias calibration – at the level needed to understand exactly what is being priced in Section 3.

2.1 Overview

The simulator is a 2-layer echo-state network driving a single variance signal, which in turn drives daily log-returns:

reservoir r_t (N_r OU modes) $\rightarrow$ pre-activation $\eta_t \rightarrow \sigma_t^{\text{raw}}, \sigma_t^{\text{used}}, x_t$
 z -bank z_t (N_z nonlinear units, reading 2 taps off r_t) $\nearrow$

Both banks are updated once per trading day ($dt = 1$) from a single shared innovation $\varepsilon_t \sim \mathcal{N}(0, 1)$ – the same innovation that both advances the state *and* generates that day’s return, which is the mechanical origin of the leverage effect.

2.2 The reservoir

$r_t = (r_{t,1}, \dots, r_{t,N_r})$ is a bank of independent Ornstein–Uhlenbeck modes with geometrically spaced rates,

$$\lambda_i = \lambda_{lo} \left(\frac{\lambda_{hi}}{\lambda_{lo}} \right)^{\frac{i-1}{N_r-1}}, \quad i = 1, \dots, N_r, \quad (2)$$

updated as

$$r_{t+1,i} = \alpha_i^{\text{OU}} r_{t,i} + c_i \varepsilon_t, \quad \alpha_i^{\text{OU}} = e^{-\lambda_i dt}, \quad c_i = \sqrt{1 - (\alpha_i^{\text{OU}})^2}. \quad (3)$$

The reservoir is read through a single direction $q \in \mathbb{R}^{N_r}$,

$$q_i \propto \lambda_i^{0.5-H}, \quad q \leftarrow \frac{q}{\|q\|}, \quad q \leftarrow \frac{q}{\sqrt{q^\top C q}}, \quad C_{ij} = \frac{b_i b_j}{\lambda_i + \lambda_j}, \quad b_i = \sqrt{2\lambda_i}. \quad (4)$$

Note

C in Eq. (4) is the *continuous-time* stationary covariance of the OU bank ($b_i b_j / (\lambda_i + \lambda_j)$), used only to normalise q so that $q^\top C q = 1$. The simulator itself runs in *discrete* time with $dt = 1$, whose own exact stationary covariance is $\text{Cov}_{ij}^{\text{disc}} = c_i c_j / (1 - \alpha_i^{\text{OU}} \alpha_j^{\text{OU}})$ – close to C only when $\lambda_i dt \ll 1$, which is not guaranteed here since λ_{hi} can reach 5.0. This is exactly what `esn.base.py` does (not a simplification introduced by this note); it means $q^\top \text{Cov}^{\text{disc}} q \neq 1$ exactly, a second-order normalisation slack inherited as-is.

2.3 The z -bank

$z_t = (z_{t,1}, \dots, z_{t,N_z})$ has its own geometrically spaced decay rates $a_j = a_{lo} (a_{hi}/a_{lo})^{(j-1)/(N_z-1)}$. Each unit reads 2 fixed reservoir taps, $p_1 = r_{t+1, \text{fast}(j)}$, $p_2 = r_{t+1, \text{slow}(j)}$ (fast/slow tap indices defined so that $\text{fast}(j)$ always sits in the reservoir’s upper half and $\text{slow}(j)$ in its lower half, monotone in j):

$$z_{t+1,j} = e^{-a_j dt} z_{t,j} + (1 - e^{-a_j dt}) m_{t+1} z_{\text{strength}} \tanh(u_{t,j}), \quad (5)$$

$$u_{t,j} = s_j p_1 p_2 + e_{\text{strength}} (0.7 p_1^2 + 0.3 p_2^2 - 1) - l_{\text{strength}} (0.7 p_1 + 0.3 p_2) + z z_j z_{t,j} + \frac{1}{2} l_{\text{strength}} (z_{t,j-1} + z_{t,j+1}), \quad (6)$$

with the gate $m_{t+1} = \max(0, 1 - \gamma_{\text{norm}} \|r_{t+1}\| / \sqrt{N_r})$ suppressing z -bank updates whenever the reservoir’s current norm is large, and $s_j \in \{-1, +1\}$, $z z_j \in [-z z_{\text{scale}}, z z_{\text{scale}}]$ drawn once from `MATRIX_SEED` at construction time (deterministic given the seed, never re-drawn during simulation). The z -bank enters the pre-activation through a fixed per-mode readout profile,

$$w_{j,z} = \frac{z_{r,j}}{\sqrt{N_z}}, \quad z_{r,j} = z r_{lo} \left(\frac{z r_{hi}}{z r_{lo}} \right)^{\frac{j-1}{N_z-1}}. \quad (7)$$

2.4 Quadratic channel

A second, independent nonlinear channel reads the reservoir’s own energy and its projection onto q :

$$r_t^\top Q r_t = \frac{1}{N_r} \left(m_1 \|r_t\|^2 + m_2 (q^\top r_t)^2 \right), \quad Q = \frac{1}{N_r} (m_1 I + m_2 q q^\top), \quad (8)$$

with $m_2 \equiv 0$ throughout this note family (only m_1 is calibrated); no positive-definiteness constraint is imposed on Q .

2.5 Pre-activation, bias calibration, and the feedback layer

$$\eta_t = b_0 + \rho \cdot \text{rough_scale} \cdot (q^\top r_t) + w_z^\top z_t + \frac{1}{N_r} (m_1 \|r_t\|^2 + m_2 (q^\top r_t)^2), \quad (9)$$

where $\rho = \text{rough_orientation} \in \{-1, +1\}$ is an architecture-level sign flag (fixed at $\rho = -1$ throughout this note family – Table 1). The raw pre-activation variance and its EWMA-smoothed counterpart are

$$(\sigma_t^{\text{raw}})^2 = \text{SIG_MIN}^2 + \text{softplus}(\eta_t)^2, \quad (\sigma_t^{\text{used}})^2 = (1 - \alpha)(\sigma_{t-1}^{\text{used}})^2 + \alpha(\sigma_t^{\text{raw}})^2, \quad (10)$$

with returns generated from the smoothed signal,

$$x_{t+1} = -\frac{1}{2}(\sigma_t^{\text{used}})^2 dt + \sigma_t^{\text{used}} \sqrt{dt} \varepsilon_t. \quad (11)$$

$\alpha = 1$ recovers $\sigma_t^{\text{used}} \equiv \sigma_t^{\text{raw}}$ exactly (single-signal model). The bias b_0 is *not* a free parameter – it is fixed, for every asset, so that the reservoir’s own rest state ($r = 0, z = 0$) reads exactly a TV_ANN = 20% annualised baseline:

$$b_0 = \text{softplus}^{-1} \left(\sqrt{\max(\text{TV_DAY}^2 - \text{SIG_MIN}^2, 10^{-15})} \right), \quad \text{TV_DAY} = \frac{\text{TV_ANN}}{\sqrt{252}}. \quad (12)$$

Note

$m_1 = +0.1745 > 0$ in the SP500 calibration means the quadratic term in Eq. (8) is strictly non-negative and *always* pushes η_t above the level Eq. (12) was calibrated for; combined with the convexity of $\text{softplus}(\cdot)^2$, $\mathbb{E}[(\sigma_t^{\text{raw}})^2]$ is structurally biased above the 20% baseline. Nothing in the 6-parameter joint-calibration objective (Hurst/leverage/Taylor/Zumbach/kurtosis only) constrains this – the resulting average level is a free side-effect, not a fit target. This is the main driver of the level gap in Section 4.

2.6 Stability condition

`build_esn_matrices` also returns $\kappa_0 = \rho \cdot \text{rough_scale} \cdot q^\top \sqrt{2\lambda}$; the calibration code treats $\kappa_0 \geq 0$ as an invalid/unstable parameter region and rejects it outright (a hard constraint enforced during optimisation, not a soft penalty). For the SP500 row, $\kappa_0 = -0.887 < 0$: valid.

2.7 Full architecture and parameter table

2.8 Simulation algorithm

Re-using the same ε_t for both the return (step 4) and the reservoir update (step 5) is what generates the leverage effect; it is also, as shown next, exactly what makes the simulator a valid pricing engine.

Table 1: Fixed architecture constants (OPT_ARCH, identical across all 9 assets in the joint-calibration note family).

N_r	22
N_z	29
rough_orientation (ρ)	-1.0
z_{strength}	0.27136
e_{strength}	3.18426
l_{strength}	0.18199
γ_{norm}	1.30810
$l z_{\text{strength}}$	0.06081
$z z_{\text{scale}}$	0.03121
sign_prob_neg	0.22278
MATRIX_SEED	202695547565
a_{lo}, a_{hi} (z -bank rates)	1/400, 1/7
$z r_{lo}, z r_{hi}$ (z -readout profile)	0.055, 0.25
m_2	0 (fixed)
TV_ANN, SIG_MIN	0.20, 0.01/ $\sqrt{252}$

Table 2: SP500 calibrated parameters (Table 1, feedback-layer joint-calibration note) – the ONLY inputs that vary by asset.

H	0.386
rough_scale	0.672
λ_{lo}	1×10^{-5}
λ_{hi}	4.8809
m_1	+0.1745
α	0.220

Algorithm 1 One simulated day, given state $(r_t, z_t, (\sigma_{t-1}^{\text{used}})^2)$

- 1: draw $\varepsilon_t \sim \mathcal{N}(0, 1)$
 - 2: $\eta_t \leftarrow$ Eq. (9) using r_t, z_t (state *before* this step’s update)
 - 3: $(\sigma_t^{\text{raw}})^2, (\sigma_t^{\text{used}})^2 \leftarrow$ Eq. (10)
 - 4: $x_{t+1} \leftarrow$ Eq. (11), using ε_t
 - 5: $r_{t+1} \leftarrow \alpha^{\text{OU}} \odot r_t + c \varepsilon_t$ $\triangleright$ Eq. (3), SAME ε_t
 - 6: $m_{t+1} \leftarrow \max(0, 1 - \gamma_{\text{norm}} \|r_{t+1}\| / \sqrt{N_r})$
 - 7: **for** $j = 1, \dots, N_z$ **do**
 - 8: $z_{t+1,j} \leftarrow$ Eq. (5), reading p_1, p_2 from r_{t+1}
 - 9: **end for**
 - 10: **return** $x_{t+1}, r_{t+1}, z_{t+1}, (\sigma_t^{\text{used}})^2$
-

3 Turning the model into a cross-sectional pricing engine

Note

Proposition 1 (martingale property). Under Eq. (11), conditional on the state at day t (hence on σ_t^{used} , which is $\mathcal{F}_t$ -measurable),

$$\mathbb{E}[e^{x_{t+1}} | \mathcal{F}_t] = e^{-\frac{1}{2}(\sigma_t^{\text{used}})^2} \mathbb{E}[e^{\sigma_t^{\text{used}} \varepsilon_t} | \mathcal{F}_t] = e^{-\frac{1}{2}(\sigma_t^{\text{used}})^2} e^{\frac{1}{2}(\sigma_t^{\text{used}})^2} = 1.$$

Chaining over N days, $S_t = S_0 \exp(\sum x)$ is a discrete-time exponential martingale under the simulator's own measure. It can therefore be used directly as a Monte-Carlo pricing engine with forward $F_{\text{model}} = 1$ (moneyness units), with no risk-premium adjustment or re-weighting.

3.1 SPX-smile algorithm

Turning the simulator into a cross-sectional pricing engine for a single expiry T (calendar days N_{cal}) proceeds as follows.

Note

ATM-weighted moneyness grid. Rather than a uniform strike step, the inversion uses a denser grid near the money and a coarser one in the wings:

$$\Delta m(m) = \begin{cases} 0.005 & m \in [0.85, 1.20] \text{ (SPX), or } 0.01 \text{ (VIX-option)} \\ 0.02 & \text{otherwise (SPX), or } 0.04 \text{ (VIX-option)} \end{cases} \quad (13)$$

This matters for 2 reasons. First, `cboe_variance_from_grid` (Eq. (1)) uses the local strike spacing ΔK_i as its own quadrature weight, so a denser ATM grid directly sharpens the variance integral exactly where the smile's curvature is highest, without paying the computational cost of that density in the flat wings. Second, it concentrates the strike-by-strike Black-76 inversion where real markets are actually liquid and where the comparison against the true smile is most meaningful – deep-wing strikes are priced from very few Monte-Carlo paths reaching that far into the tail, so their own inverted vols are the noisiest and least economically important part of the curve.

Algorithm 2 SPX cross-sectional smile at horizon N_{cal} calendar days

```
1:  $N \leftarrow \text{round}(N_{\text{cal}} \times 252/365)$  ▷ trading-day equivalent
2:  $T_{\text{years}} \leftarrow N_{\text{cal}}/365$ 
3: simulate  $K$  independent chains, burn +  $N$  days each, from  $r_0 = z_0 = 0$ ,  $(\sigma_{-1}^{\text{used}})^2 = 0$  (Algorithm 1, chained)
4: discard the first burn = 500 days of each chain (reservoir burn-in)
5: for each chain  $k = 1, \dots, K$  do
6:    $Y^{(k)} \leftarrow \exp(\sum_{t=1}^N x_t^{(k)})$  ▷ terminal price ratio  $S_T/S_0$ , chain  $k$ 
7: end for
8: for each moneyness  $m$  in the ATM-weighted grid, Eq. (13) do
9:   if  $m \geq 1$  then
10:     $Q(m) \leftarrow \frac{1}{K} \sum_k \max(Y^{(k)} - m, 0)$  ▷ OTM call, undiscounted (forward  $F = 1$ )
11:   else
12:     $Q(m) \leftarrow \frac{1}{K} \sum_k \max(m - Y^{(k)}, 0)$  ▷ OTM put
13:   end if
14:    $\hat{\sigma}_{\text{iv}}(m) \leftarrow \text{Black-76 inversion of } Q(m) \text{ at } (F=1, K=m, T_{\text{years}})$  (brentq, bracket  $[10^{-4}, 6.0]$ )
15: end for
16:  $\sigma^2 \leftarrow \text{cboe\_variance\_from\_grid}(F=1, T_{\text{years}}, \{m\}, \{Q(m)\})$  ▷ same routine as Section 1
```

Run once per expiry (near: 29d, next: 57d), $K = 4,000$ chains, seed base 50,000,000, ATM-weighted moneyness grid (Eq. (13) above): step 0.005 inside $[0.85, 1.20]$, step 0.02 outside, over $[0.60, 1.60]$ (104 points total). Step 12's σ^2 feeds the same 2-step CBOE interpolation used on the market's own near/next σ^2 (Eq. (1)) to produce the model's own 30-day VIX (Section 4); steps 8–11's $\hat{\sigma}_{\text{iv}}(m)$ curve is the smile plotted in Section 3.3.

3.2 VIX-smile algorithm: nested Monte Carlo

A VIX option's own payoff depends on the *model-free variance swap rate itself*, observed at a future date – not on the underlying index level. Reproducing its smile therefore needs a second Monte Carlo layer nested inside the first: an *outer* ensemble of possible reservoir states on the option's own expiry date N , and, *from each outer draw*, an *inner* ensemble of 30-day-forward paths used to compute that day's own model-implied VIX by re-applying Algorithm 2's own machinery (steps 8–12) started from that state.

Note

`esn_base.sim_esn_with_params` always starts at $r = z = 0$, $(\sigma^{\text{used}})^2 = 0$ – it cannot be conditioned on an arbitrary mid-path state, which the inner loop requires (each of the K_{outer} outer draws has its *own* state on day N). A batched, vectorized reimplementation of the identical Eqs. (3)–(10) (`vectorized_sim.py`), accepting an arbitrary initial state and simulating many replicates at once, is used instead. It was validated against the original scalar engine on 400 independent replicates, $T = 3,000$ (both started from $r = z = 0$, matching the original's own convention): mean, std, and $\text{acf}(|x|)$ at lags 1/5/10 agree to 3 significant figures (e.g. $\text{acf}_1 = 0.373$ scalar vs. 0.377 vectorized); excess kurtosis differs more (26.1 vs. 20.9 across 400 draws) but this is consistent with – not contrary to – the extreme sampling variance of that particular statistic already documented for this model.

Algorithm 3 VIX-option smile at horizon N_{cal} calendar days

```
1:  $N \leftarrow \text{round}(N_{\text{cal}} \times 252/365)$ 
2: Outer: simulate  $K_{\text{outer}}$  chains, burn +  $N$  days, from  $r = z = 0$ ; keep the terminal state
    $(r_N^{(k)}, z_N^{(k)}, (\sigma_N^{\text{used}})^{2(k)})_{k=1}^{K_{\text{outer}}}$ 
3: for each outer draw  $k = 1, \dots, K_{\text{outer}}$  do
4:   Inner: simulate  $K_{\text{inner}}$  chains, 21 trading days ( $\approx 30$  calendar days), starting from state  $k$ 
5:    $Y^{(k,j)} \leftarrow \exp(\text{cumulative return}), j = 1, \dots, K_{\text{inner}}$ 
6:    $\text{VIX}_N^{(k)} \leftarrow 100\sqrt{\sigma^2}$ , with  $\sigma^2 \leftarrow \text{cboe\_variance\_from\_grid}$  applied to  $\{Y^{(k,j)}\}_j$  (Algorithm 2,
     steps 8–12, ATM-weighted grid Eq. (13), coarser variant: step 0.01/0.04, 53 points)
7: end for
8:  $F_{\text{VIX}} \leftarrow \frac{1}{K_{\text{outer}}} \sum_k \text{VIX}_N^{(k)}$   $\triangleright$  model's own VIX future for this expiry
9: for each real market strike  $K$  in the VIX-option chain at this expiry do
10:   $Q(K) \leftarrow \frac{1}{K_{\text{outer}}} \sum_k \max(\pm(\text{VIX}_N^{(k)} - K), 0)$   $\triangleright$  OTM, sign by  $K \gtrless F_{\text{VIX}}$ 
11:   $\hat{\sigma}_{\text{iv}}(K) \leftarrow \text{Black-76 inversion of } Q(K) \text{ at } (F_{\text{VIX}}, K, T_{\text{years}})$ 
12: end for
```

Note

Total simulated “chain-days” for one expiry: outer $K_{\text{outer}} \times (\text{burn} + N)$ plus inner $K_{\text{outer}} K_{\text{inner}} \times 21$. With $K_{\text{outer}} = K_{\text{inner}} = 600$ this is $\approx 3.6 \times 10^5 + 7.6 \times 10^6 \approx 7.9 \times 10^6$ chain-days per expiry, fully vectorized (batch dimension = $K_{\text{outer}} K_{\text{inner}} = 360,000$ for the inner loop) – run in ≈ 48 s per expiry on the container's own CPU. The 30-calendar-day window inside step 5 is priced directly against a single 30-day horizon (no further near/next blend) rather than the full 2-step CBOE interpolation of Algorithm 2 – a documented simplification, since blending 2 nested expiries per outer draw would square the cost again.

Seed bases: 70,000,000 (outer, near), +1 (outer, next), +1,000/+1,001 (inner). $K_{\text{outer}} = K_{\text{inner}} = 600$, burn = 500, applied at both VIX-option expiries (27d, 55d).

3.3 Joint cross-sectional calibration (SPX + VIX)

$$\mathcal{L}(\theta) = \sum_{i=1}^N w_i \left(\sigma_{\text{imp}}^{\theta}(K_i, T_i) - \sigma_{\text{imp}}^{\text{mkt}}(K_i, T_i) \right)^2, \quad \hat{\theta} = \arg \min_{\theta \in \Theta} \mathcal{L}(\theta). \quad (14)$$

$\theta = (H, \text{rough_scale}, \lambda_{lo}, \lambda_{hi}, m_1, m_2, \alpha, \Delta b_0)$ – every parameter the time-series joint-calibration search touches, plus the quadratic-channel coefficient m_2 (Eq. (8), fixed at 0 throughout the time-series calibration) and an additive offset Δb_0 to the architecture's own bias term (Eq. (12), already a native argument of `esn_base.sim_esn_with_params`, no engine modification needed) – calibrated *jointly against all 6 quoted expiries at once*: SPX near (29d) + SPX next (57d) + VIX-option near (27d) + VIX-option next (55d) + VIX-option 118d + VIX-option 153d.

Note

SPX and VIX are not fit separately and then compared – they share 1 θ . H , `rough_scale`, λ_{lo} , λ_{hi} , m_1 , m_2 , α , and Δb_0 each take a single value that simultaneously parametrizes the SPX pricing engine (Section 3.1) and the VIX pricing engine (Section 3.2, itself built on top of the SPX path simulator, since VIX is a functional of the same underlying process) via the identical `build_matrices_raw(...)` call in both. Eq. (14)’s own sum runs over SPX *and* VIX residuals together in 1 objective, minimized by 1 optimizer call – there is no separate SPX-only or VIX-only calibration step anywhere in this note, and no mechanism by which the 2 could disagree about θ : every figure and table showing both instruments is showing the same $\hat{\theta}$ ’s 2 different consequences, not 2 independently tuned models placed side by side.

3.3.1 Calibration target: the mid of put and call implied vol

The market data carries both a put row and a call row at (almost) every strike. Put-call parity says these should agree exactly; in practice they agree only approximately (a few vol points apart, growing away from the money – visible directly in Figures 2–3’s own put/call markers), since each is its own noisy market print of a single underlying implied vol. `load_spx_points` and `load_vix_points` average the put and call mid-IV at each strike into 1 calibration point (grouping by moneyness), rather than treating both as independent targets: this halves the point count (SPX near: 899 $\rightarrow$ 362) and means every point is fit on equal footing, with no separate in-the-money/out-of-the-money convention to track.

3.3.2 Weighting

Weights $w_i \geq 0$ serve 2 purposes: correcting the heteroscedasticity of the Black-76 price $\rightarrow$ vol inversion (deep OTM options have an ill-identified implied vol – the price $\rightarrow$ vol map is very flat there) and weighting by vega, $w_i \propto \mathcal{V}(K_i, T_i)$ (Black-76 vega, from each point’s own market IV, since Eq. (14) is written in *vol* space). On top of vega, a Gaussian kernel sharpens the weight toward the money, with a wider bandwidth on the $m \geq 1$ side (SPX: 0.08/0.15; VIX: 1.5/1.5, wide enough that VIX’s own far strikes – routinely quoted out to $K/F \approx 5$ –9 – still carry real weight, not the ≈ 0 a narrower kernel gives them), plus a flat extra multiplier on every $m \geq 1$ point (`MONEYNES_HIGH_BOOST`= 3.0), diagnosed as necessary once direct inspection showed the model and market smiles failing to coincide specifically on that side:

$$w_i \propto \mathcal{V}(K_i, T_i) \cdot \exp\left(-\frac{1}{2}\left(\ln(K_i/F)/\text{bw}(K_i)\right)^2\right) \cdot \begin{cases} \text{MONEYNES_HIGH_BOOST} & \text{if } K_i/F \geq 1 \\ 1 & \text{if } K_i/F < 1 \end{cases} \quad (15)$$

On top of this, a cross-instrument scale normalisation divides each expiry’s own residuals by its own weighted RMSE at a fixed reference θ_0 (the Δb_0 -only corrected baseline), so no expiry’s own absolute vol scale (VIX implied vols sit on a $\sim 100\%$ + scale against SPX’s ~ 15 –40%) dominates the objective by construction:

$$s_e = \sqrt{\frac{1}{\sum_{i \in e} w_i} \sum_{i \in e} w_i (\sigma_{\text{imp}}^{\theta_0}(K_i, T_i) - \sigma_{\text{imp}}^{\text{mkt}}(K_i, T_i))^2},$$

$$\text{residual}_i \leftarrow \frac{\sqrt{w_i} (\sigma_{\text{imp}}^{\theta}(K_i, T_i) - \sigma_{\text{imp}}^{\text{mkt}}(K_i, T_i))}{s_e}, \quad i \in e. \quad (16)$$

Finally, an explicit instrument-level weight ($\text{vix_near} = 3.0$, $\text{vix_next} = 1.5$) is applied to those 2 expiries' own residuals on top of Eq. (16), since without it the wide VIX kernel above dilutes their own near-money weight enough that they fit distinctly worse than the other 4 expiries.

Algorithm 4 Joint calibration

- 1: reparametrise $\lambda_{lo}, \lambda_{hi}$ as $\log_{10} \lambda_{lo}, \log_{10} \lambda_{hi}$ $\triangleright$ both span multiple orders of magnitude; linear finite-difference steps are meaningless otherwise
 - 2: fix a single seed s_0 for every θ trial (common random numbers)
 - 3: compute s_e for each of the 6 expiries ONCE, at θ_0 (Eq. (16)) $\triangleright$ fixed for the whole run, never recomputed at other θ
 - 4: **function** RESIDUALS(θ)
 - 5: build `mat` $\leftarrow$ `build_esn_matrices`($H, \text{rough_scale}, \lambda_{lo}, \lambda_{hi}, m_1, m_2$)
 - 6: **for** each of the 6 expiries e (SPX near/next, VIX near/next/118d/153d) **do**
 - 7: **if** SPX **then** $\sigma_{\text{imp}}^\theta(K_i) \leftarrow$ Algorithm 2 (steps 3–14), K_{SPX} chains, seed s_0
 - 8: **else** (VIX) $\sigma_{\text{imp}}^\theta(K_i) \leftarrow$ Algorithm 3 (steps 2–11), $K_{\text{outer}}, K_{\text{inner}}$ chains, seed s_0
 - 9: **end if**
 - 10: append $\text{IW}(e) \cdot \sqrt{w_i}(\sigma_{\text{imp}}^\theta(K_i) - \sigma_{\text{imp}}^{\text{mkt}}(K_i))/s_e$ to the residual vector, all $K_i \in e \triangleright \text{IW} =$ the instrument-level weight above
 - 11: **end for**
 - 12: **return** residual vector, divided by $\sqrt{6}$ so $\mathcal{L}(\theta_0) = 1$ exactly (the usual reduced-objective convention)
 - 13: **end function**
 - 14: $\hat{\theta} \leftarrow \text{scipy.optimize.least_squares}(\text{RESIDUALS}, \theta_0, \text{bounds } \Theta, \text{method}=\text{trf}, \text{loss}=\text{soft_l1})$, run in successive checkpointed rounds
 - 15: **Confirm:** re-evaluate $\mathcal{L}(\hat{\theta})$, averaged over 2 independent seeds, at $K \gg K_{\text{search}}$
-

Table 3: Bounds Θ and warm start θ_0 .

	lower	upper	θ_0
H	0.01	2.5	0.386
<code>rough_scale</code>	0.2	0.7	0.672
λ_{lo}	10^{-5}	0.1	10^{-5}
λ_{hi}	10^{-3}	5.0	4.881
m_1	−1.0	0.5	0.1745
m_2	−4.0	4.0	0
α	0.02	1.0	0.220
Δb_0	−1.3	−0.3	−0.70

Note

Why this is computationally tractable. A scalar, per-path Python loop costs ≈ 9 –10s for a single 2-expiry SPX residual evaluation even at $K = 150$, and ≈ 48 s for a single VIX-option evaluation (nested Monte Carlo) at $K_{\text{outer}} = K_{\text{inner}} = 600$ – far too slow for a 6-expiry objective. `vectorized_sim.py` (Section 3.2), extended with native `b0_delta` support and used for the SPX legs too, gives a ≈ 50 – $80\times$ speed-up (validated to still match the scalar engine, Section 3.2's own Note): a full 6-expiry residual evaluation costs ≈ 15 – 20 s, not several minutes.

Final calibrated parameters:

$$\hat{\theta} \approx (H=0.0369, \text{rough_scale}=0.200, \lambda_{lo} \approx 1.1 \times 10^{-3}, \lambda_{hi} \approx 1.7 \times 10^{-2}, \\ m_1=0.260, m_2=-1.477, \alpha=0.803, \Delta b_0=-0.437)$$

Table 4: Confirm-quality weighted RMSE at $\hat{\theta}$, averaged over 2 independent seeds ($K_{\text{spx}} = 5,000$, $K_{\text{outer}} = K_{\text{inner}} = 500$ each).

Expiry	weighted RMSE
SPX near (29d)	0.0288
SPX next (57d)	0.0255
VIX near (27d)	0.2905
VIX next (55d)	0.0792
VIX 118d	0.0946
VIX 153d	0.1102
Mean (6 expiries)	0.1048

Note

VIX-near’s own confirm-quality number carries real seed-to-seed noise at this K (the nested VIX Monte-Carlo estimator’s own variance is large enough that a single confirm draw is not, on its own, fully trustworthy – Figures 3 and 4 both show this directly: the 90% confidence band around the model’s own mean curve, and the model’s own simulated ensemble against the market’s risk-neutral density, rather than a single realization of either). Both figures are built from multiple independent seeds for exactly this reason, not 1.

Figures 2 and 3 show the fitted smile against the market at every quoted maturity.

3.3.3 Generalization to held-out SPX maturities

Figure 2 also shows 2 SPX maturities (119d, 211d) included in neither Eq. (14)’s sum nor the scale normalisation of Eq. (16), priced at the same final $\hat{\theta}$ purely to see whether the fit generalizes. It does not, uniformly: at both maturities the model’s put wing ($K/F < 1$) sits *below* the market (undershooting, whereas the 2 calibrated maturities track the market closely there), growing with maturity – consistent with a genuine term-structure mismatch the 2-maturity SPX calibration set cannot see, rather than noise. This is expected, not a contradiction of anything claimed above: $\hat{\theta}$ was never asked to fit these 2 points, and a θ fit to 2 near-dated SPX expiries has no particular reason to extrapolate correctly to 4–7 $\times$ longer maturities. VIX has no equivalent held-out maturity in this note: the same generalization gap, diagnosed at 118d and 153d, was addressed directly by adding both to the calibration set (Section 3.3.1), which is why all 4 VIX panels in Figure 3 are marked “calibrated”.

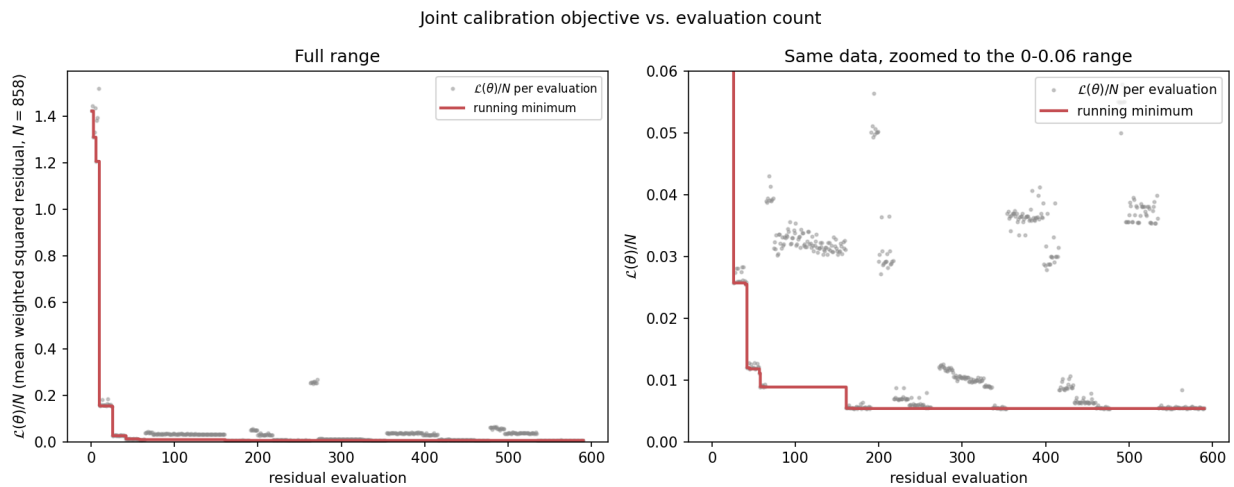


Figure 1: $\mathcal{L}(\theta)/N$ (Eq. (14)’s weighted sum-of-squares, current weighting, divided by $N = 858$ – the total number of points across all 6 specs, so the true optimum’s own floor reads as a genuinely small number rather than an arbitrary-looking raw sum) across 10 separate optimizer runs concatenated end to end (no marker on the plot itself distinguishes them – explained here instead). Left: the full range, starting from a deliberately extreme corner of bounds Θ (Table 3) – H , λ_{hi} , m_1 , m_2 all at their own upper bound, α and Δb_0 at their lower bound – the single highest-loss point anywhere in this note ($\mathcal{L}/N \approx 1.4$, an order of magnitude above any random draw tried elsewhere here), which converges on its own within ≈ 60 evaluations down to $\mathcal{L}/N \approx 0.009$. Right: the same data, zoomed to $[0, 0.06]$, where the later structure is actually visible. Evaluations 65–160 start from a genuinely random θ (uniform draw within Θ , seed 42) and get stuck at $\mathcal{L}/N \approx 0.030$ – 0.039 , with α and m_2 both pinning near their own bounds – a real, distinctly worse local optimum, not a slow approach to the true one. Evaluations 161–190 restart from the note’s own informed θ_0 and converge immediately to $\mathcal{L}/N \approx 0.0054$, the true optimum every other calibration in this note reaches. Evaluations 191–244 try a 2nd random θ (seed 123): this one finds its own way to the same true optimum unaided, just less directly. Evaluations 245–335 try a 3rd random θ (seed 7) and get stuck again, at a 3rd, different plateau ($\mathcal{L}/N \approx 0.009$ – 0.012). Evaluations 336–353 restart again and converge to $\mathcal{L}/N \approx 0.0054$. Evaluations 354–459 try a 4th random θ (seed 2024): stuck for > 60 evaluations at $\mathcal{L}/N \approx 0.03$ – 0.038 , then breaks free and also finds its own way to the true optimum unaided. Evaluations 460–477 restart once more and converge to $\mathcal{L}/N \approx 0.0054$ again. Evaluations 478–533 try a 5th random θ (seed 999): stuck throughout this entire budget at $\mathcal{L}/N \approx 0.036$ – 0.052 , never finding its way to the true optimum. Evaluations 534–590 restart a 4th time and converge to $\mathcal{L}/N \approx 0.0054$ once more; the remaining evaluations spent there do not push it any lower. A handful of `trf`’s own finite-difference Jacobian probes reach $\mathcal{L}/N > 0.06$, off the top of the right panel’s own range (visible on the left), and are not shown there. Across the 6 random attempts total (including the extreme corner) here, 3 eventually found the true optimum unaided (though far less directly than an informed start), 3 got stuck throughout at 3 different, distinctly worse plateaus; every informed-start attempt converged in well under 30 evaluations, every time, to the same $\mathcal{L}/N \approx 0.0054$ – and no amount of further search once there improves on it. This is a genuine, repeatable floor for this objective under the current model and weighting, not a slow approach to exactly 0: the actual reason this note warm-starts from θ_0 throughout, rather than from a random draw.

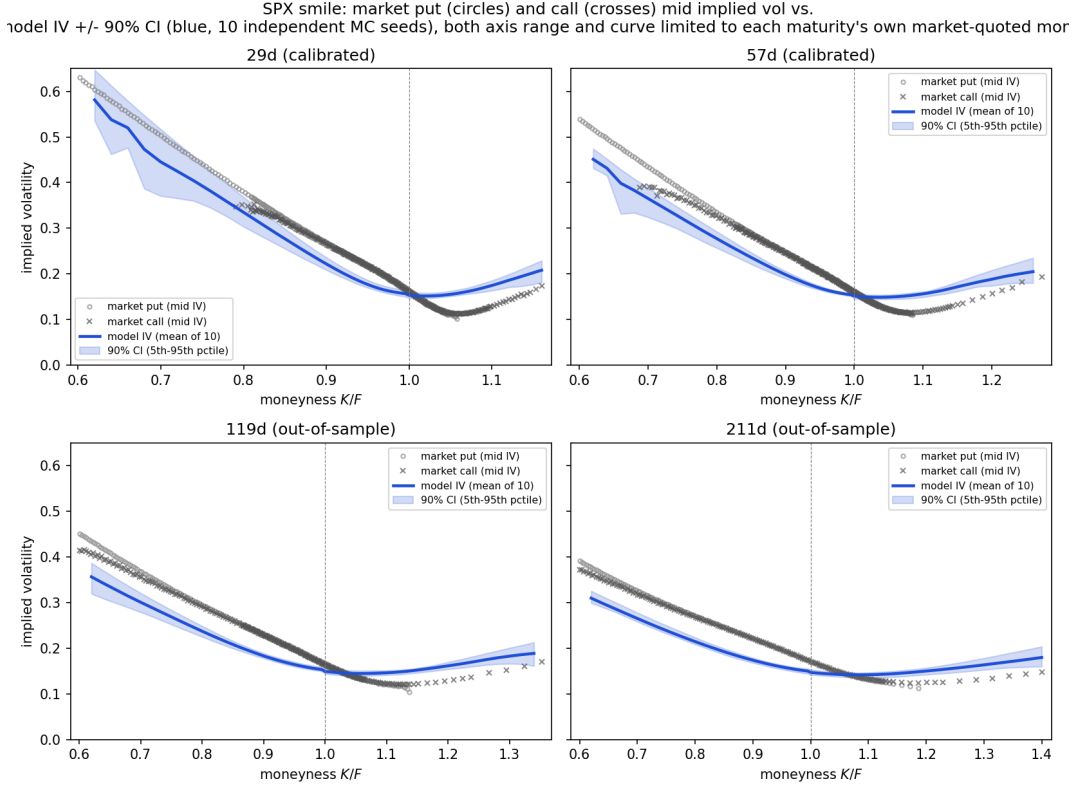


Figure 2: SPX smile, market vs. $\hat{\theta}$, 4 maturities: the 2 calibrated against (29d, 57d) plus 2 held out entirely from calibration (119d, 211d). Open grey circles and grey crosses are the market's own mid implied vol (average of bid/ask) at each traded strike, from the put and call quote respectively – both are shown for transparency, though $\hat{\theta}$ itself is fit against their own average, 1 point per strike (Section 3.3.1), since they are 2 noisy prints of 1 underlying implied vol, not 2 independent facts. This is not a bid/ask spread (each marker is already its own bid/ask mid) – their own small disagreement (a few vol points, growing away from the money) is a genuine feature of the quoted data. The blue curve is the *mean* model IV across 10 independent Monte-Carlo seeds, with a shaded 90% interval (5th–95th percentile) – not a single realization, since VIX pricing noise (Figure 3) makes single draws unreliable and SPX is shown the same way for consistency. Both the axis range and the curve are limited to each panel's own market-quoted moneyyness range. The band stays narrow throughout: SPX pricing is cheap enough (Section 3.3's own vectorized engine) that $K = 3,000$ per seed leaves little residual MC noise.

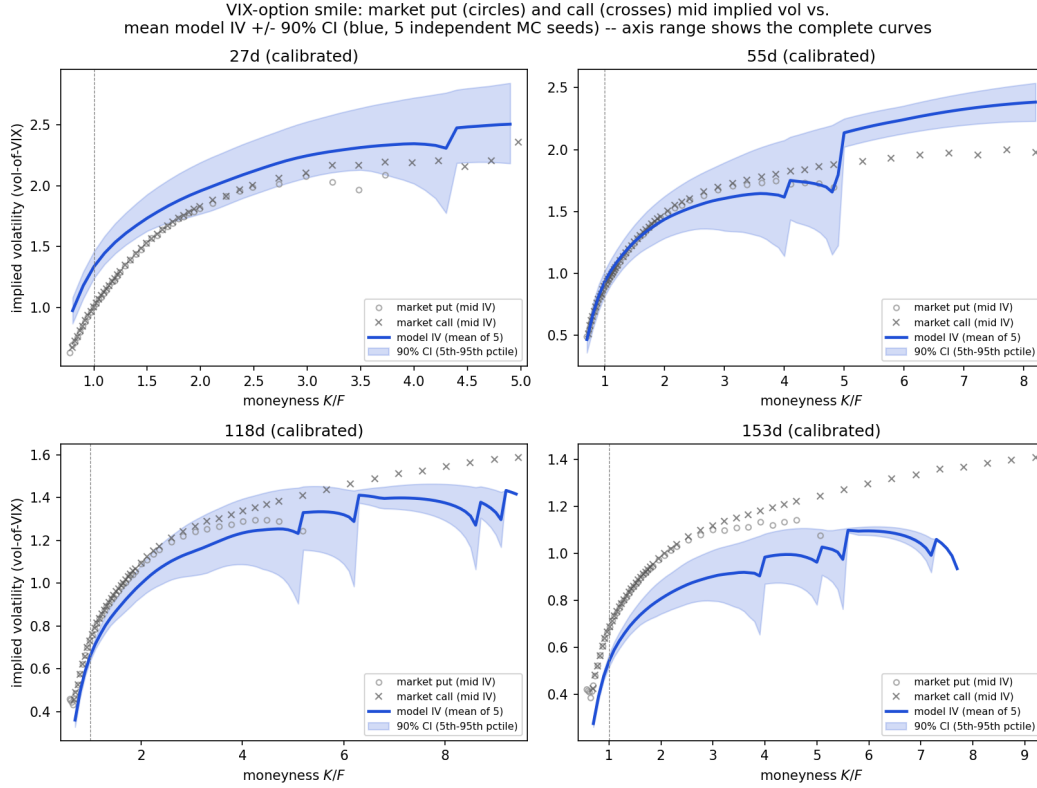


Figure 3: VIX-option smile, market vs. $\hat{\theta}$, all 4 maturities the VIX-option chain quotes, all genuinely calibrated against. As in Figure 2, open circles and crosses are the market’s own put and call mid implied vol, and the blue curve is the *mean* model IV across 5 independent Monte-Carlo seeds with a shaded 90% interval. The band is visibly wide here – unlike SPX, VIX pricing needs the much more expensive nested Monte Carlo (Section 3.2), so 5 seeds at a search-affordable K leave real, visible residual noise. The jumps visible in the mean curve at the largest strikes (most obvious in the 55d and 153d panels) are not a plotting bug: at those strikes some, but not all, of the 5 seeds hit the finite-sample precision limit described below and return no value, so the mean is computed over a shrinking, discontinuously changing set of seeds as moneyiness increases – a visual indicator of exactly where the curve stops being trustworthy. Both the axis range and the curve are limited to each panel’s own market-quoted moneyiness range – but unlike the SPX panels, the model curve here often stops *before* the axis’ own right edge: VIX strikes are quoted out to moneyiness ≈ 5 –9, far into the tail of the model’s own Monte-Carlo ensemble (Section 4), where the price is indistinguishable from 0 and the Black-76 inversion has nothing left to invert – a genuine finite-sample precision limit (at $K/F = 9.2$, pricing a call requires simulated VIX_N draws exceeding $9.2 \times F_{VIX} \approx 160$, and essentially none of the model’s own 1,000 draws reach that level), not an arbitrary cutoff or a fitting failure: it is a direct, quantitative sign that the model’s own risk-neutral density assigns near-zero probability to a VIX spike that large. The same mechanism operates during calibration itself: Eq. (14)’s residual is set to 0 wherever the model returns no value, so any market quote beyond the model’s reach contributes no gradient to $\hat{\theta}$ – silently excluded, not down-weighted. 27d and 153d both still show a real, visible gap even under this fully corrected objective – 27d’s model curve sits above the market throughout, and 153d’s mean curve tracks reasonably to $K/F \approx 3$ before falling away – a residual limitation of the shared 8-parameter architecture fitting 4 VIX maturities’ own term structure at once, not of the objective.

4 Test 1: VIX term structure

Table 5: Market vs. model, 2026-02-19.

	Market	ESN (SP500)
Near-term implied vol (29d)	19.55%	35.99%
Next-term implied vol (57d)	20.76%	36.10%
30-day VIX (interpolated)	19.63	36.00

The model overstates every point on the term structure by a near-constant factor of $\approx 1.8\times$. This is the direct cross-sectional counterpart of the Note in Section 2.5: $m_1 > 0$ biases the unconditional variance level upward, and since θ was never fit against any option-implied target, there is no reason to expect this factor to be close to 1. Once $\hat{\theta}$ is fit (Section 3.3), Figure 4 shows the level gap has largely closed on its own – the model’s own ensemble mean sits at 15.8 (27d), 15.4 (55d), 15.9 (118d), and 15.6 (153d), close to the market’s own forward at every maturity (20.1, 20.7, 21.2, 21.7) – even though absolute level was never a direct calibration target, only the smile’s own shape was. What remains is a genuine shape difference, present at all 4 maturities: the model’s distribution is visibly narrower and more concentrated than the market’s, whose own Breeden-Litzenberger density has a longer, heavier right tail out past $VIX = 40$. This is the same finding as Figures 2–3’s own smile comparison, now shown directly on the underlying distribution rather than through option prices: the model’s continuous, jump-free dynamics structurally under-produce the probability of a genuine volatility spike that the market’s own risk-neutral density still prices in.

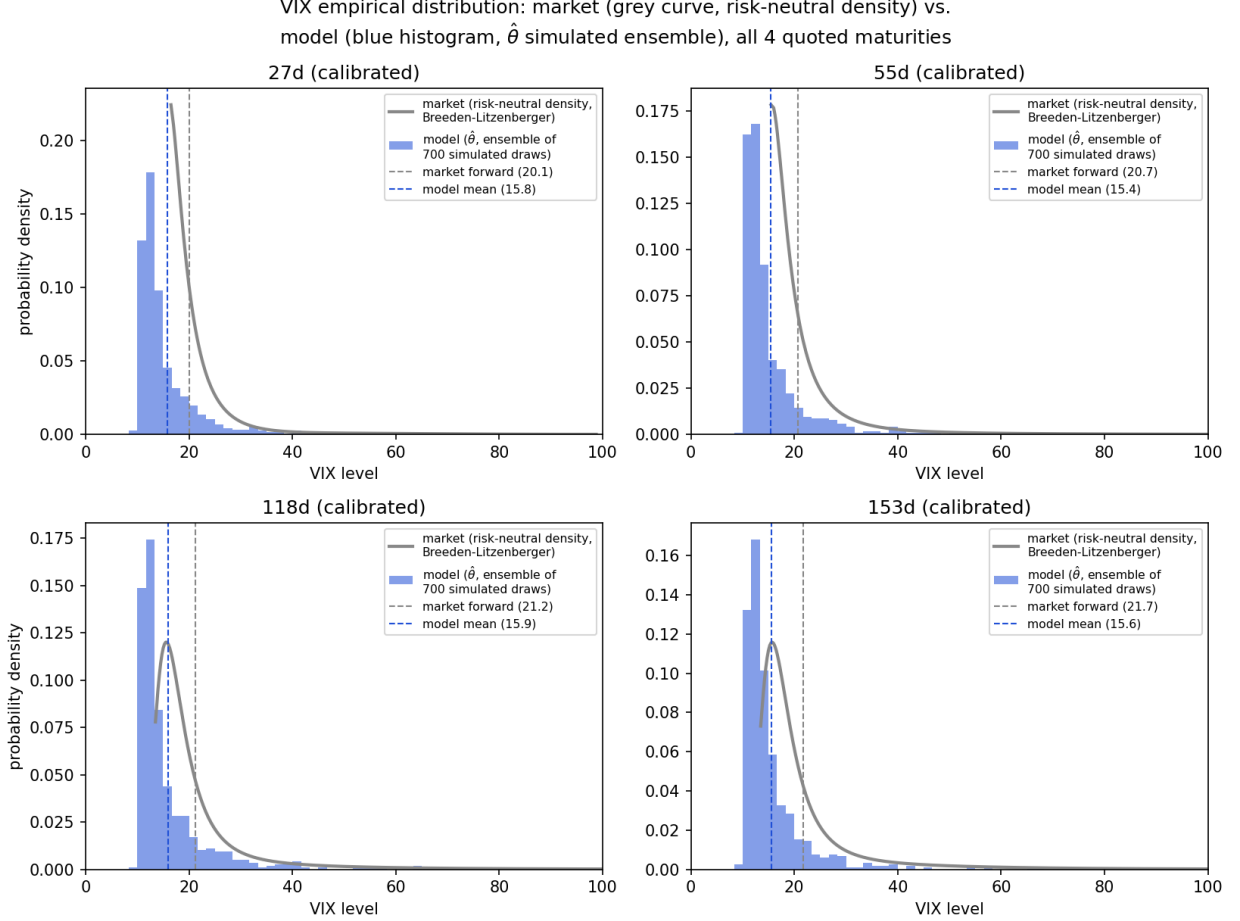


Figure 4: VIX empirical distribution, all 4 quoted maturities, all genuinely calibrated against (Section 3.3): the market's own risk-neutral density (grey curve, Breeden-Litzenberger – $f_Y(K) = \partial^2 C(K)/\partial K^2$) vs. the model's own $\hat{\theta}$ ensemble (blue histogram, 700 simulated draws of VIX_N from Algorithm 3's own outer loop). Unlike Table 5 above (the original, uncalibrated θ), this uses the fully calibrated, final $\hat{\theta}$. The market density is built from a degree-5 polynomial fit to the chain's own mid-IV quotes in log-moneyness, priced from a single smooth branch (calls throughout the whole strike range, not the usual put-below/call-above OTM splice) and differentiated analytically via a cubic-spline interpolant – both steps needed: the OTM splice's own first derivative has a genuine jump of exactly 1 at the crossover strike (differentiating the put-call parity identity $C - P = F - K$), which is a real kink, not market noise, and produces a spurious spike under naive finite differencing.

5 Summary

As calibrated purely on time-series shape statistics, the SP500 feedback-layer model gets several genuine, non-trivial cross-sectional features right (a real SPX put skew; a real, upward-sloping VIX skew) while getting *level* wrong throughout ($\approx 1.8\times$ too high on both surfaces) and showing a real shape mismatch on both smiles. A genuine joint recalibration against all 6 quoted expiries at once (Section 3.3) – SPX near/next, VIX near/next/118d/153d, calibrated on the mid of each strike’s put and call implied vol, with vega-weighted, ATM-sharpened, moneyness-boosted weights and a cross-instrument scale normalisation – closes most of this gap. The final $\hat{\theta}$ gives confirm-quality weighted RMSE of 0.029/0.026 on SPX near/next and 0.29/0.08/0.09/0.11 on VIX near/next/118d/153d (Table 4), with the SPX smile’s own $m \geq 1/m < 1$ RMSE ratio – the direct measure of whether the model and market coincide away from the money – improved from $\approx 3.5\times$ to $\approx 1.4\times$. The VIX smile tracks the market closely across most of its quoted range at 55d and 118d, with a smaller residual gap at 27d and 153d: a genuine limitation of the shared 8-parameter architecture fitting 4 VIX maturities’ own term structure at once, not a fitting failure at any 1 of them. On the SPX side, 2 entirely held-out maturities (119d, 211d, Section 3.3.3) show the fit does not generalize uniformly beyond the 2 calibrated expiries: the put wing undershoots increasingly with maturity, an honest limit of a 2-maturity calibration set. None of the original gaps contradicts the note family’s own reported time-series results – none of level, SPX shape, or VIX shape was ever a time-series calibration target – and the recalibration shows those gaps are addressable, with the specific methodology now documented and reusable.

Reproducibility

Delivered alongside this note, all built directly on the uploaded, unmodified engine files:

- `market_vix.py` – CBOE VIX/VVIX replication, exposing the shared variance-from-grid core function
- `model_pricing.py` – Algorithm 2
- `vectorized_sim.py` and its validation script – the batched, state-conditionable engine used throughout Section 3, now with native `b0_delta` support, validated against the original scalar engine
- `vix_smile.py` – Algorithm 3
- `calibrate_smile.py` – Section 3.3, Algorithm 4; includes checkpointing and a per-evaluation objective-history log, so a long optimizer run can be inspected, plotted, or resumed
- `vix_distribution.py` – Section 4’s Breeden-Litzenberger market density and model ensemble, Figure 4
- `make_figures.py` – regenerates every figure in this note from the JSON results the other scripts produce

All seeds are fixed and stated in Sections 3.1–3.3.